

Lost in the Interface: How Structured UI Complexity Challenges Large Vision Language Models in Games

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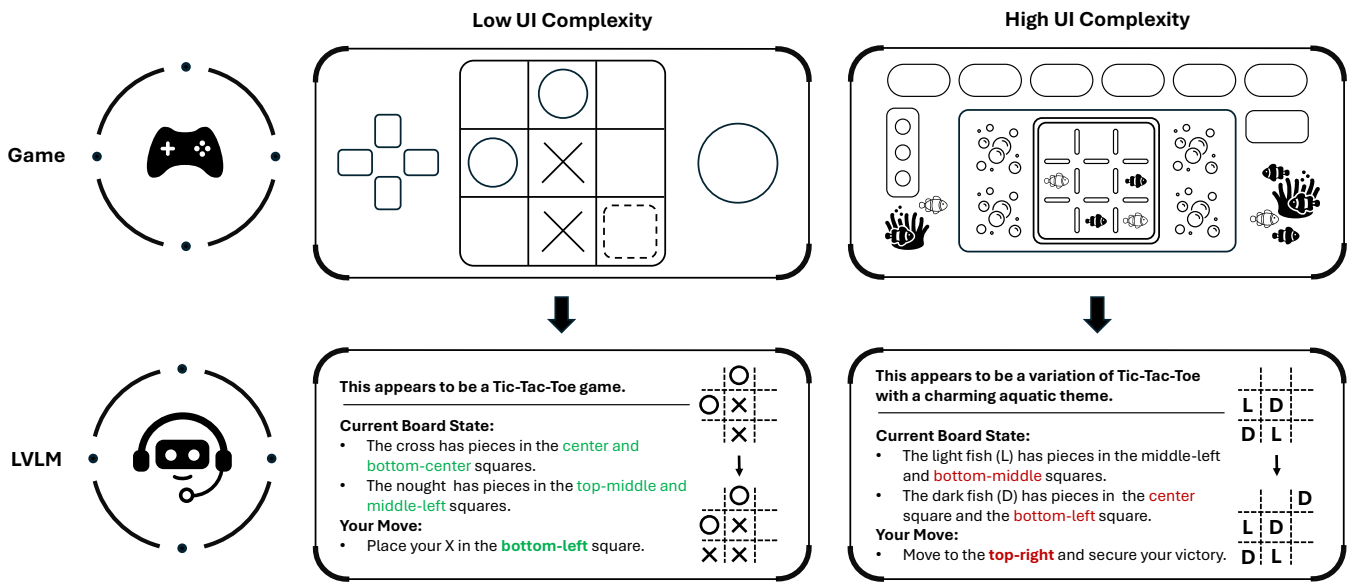


Figure 1: Large Vision-Language Models (LVLNs) show different accuracy in obtaining visual information from game screenshots when facing different degrees of UI complexity. This can reduce the accuracy of the model’s predictions for the next action and may lead to game failure.

Abstract

Current Large Vision-Language Models (LVLNs) have struggled to interpret complex game user interfaces (UIs). LVLNs frequently failing to distinguish core gameplay components from irrelevant elements. These insights highlight both the limitations of existing LVLNs and the importance of incorporating UI complexity as a key factor in evaluation. To overcome this limitation, we introduce the first benchmark for evaluating the performance of LVLNs against UI complexity across different games. We consider this work as a stepping stone toward building more UI aware

LVLNs and enabling domain-specific evaluations that better capture real-world challenges. Code and data are publicly available at <https://github.com/XiangLiSky/Lost-in-the-interface>

CCS Concepts

• **Computing methodologies** → **Activity recognition and understanding**; *Scene understanding*; Intelligent agents.



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1 Introduction

Rapid advancements in Large Vision-Language Models (LVLMs) have demonstrated their impressive capabilities in comprehending and generating multimodal content [Anthropic. 2025; Bai et al. 2023; Hurst et al. 2024; Team et al. 2023]. Specifically, LVLMs show significant potential in interactive tasks, which are becoming more sophisticated. Interactive game environments are characterized by rich visual information, diverse interaction modalities, and clear and fixed rule sets. Games serve as ideal platforms for evaluating and enhancing the multimodal abilities of LVLMs. At the same time, extant research has begun to explore the general performance of LVLMs across various gaming scenarios [Wang et al. 2025; Zheng et al. 2025]. However, a systematic investigation into how the fundamental visual complexity of game User Interface (UI) impacts the perceptual and decision-making performance of LVLMs remains underexplored. As shown in Fig. 1, we can confirm that the complexity of the UI affects LVLM's judgment during the process. Thus, we need a thorough understanding of this correlation to deploy LVLMs more widely in a range of visually intensive interactive applications.

This research aims to investigate the relationship between the visual complexity of game UI and the performance of LVLMs. The central research question that we address is: how multimodal aspects of game UI complexity influence an LVLM's accuracy of decisions regarding the next move in the game. For instance, an LVLM is faced with a task to manage a unit in a complex card game. The model may be presented with a screen filled with numerous small icons representing various card costs, unit abilities, and resource counters, competing for attention. To execute a critical action, such as activating a special ability, the LVLM has to navigate a series of pop-up menus whose layout changes dynamically. Then it has to precisely click a small, temporarily available button as well. To systematically investigate such scenarios, we need to define and employ a set of quantifiable benchmarks. These benchmarks will aim to capture the UI's complexity by measuring.

Subsequently, corresponding experiments will be conducted with representative LVLMs and game environments that show a clear level of UI complexity. By comparing the performance of these LVLMs across these varied environments, we intend to clarify the specific impact patterns and potential performance bottlenecks imposed by visual UI complexity on current-generation LVLMs. The objective is to provide initial quantitative insights into this underexplored field. We believe that it will inform the future design of AI systems that are better able to handle complex visual interfaces. Furthermore, it can guide the deployment of existing LVLMs in complex environments.

The contributions of our research are summarized as follows.

- (1) **We introduce the first benchmark and dataset for evaluating LVLM against game UI complexity:** We have constructed a curated dataset with different levels of game UI complexity. This dataset is accompanied by new evaluation metrics that enable systematic measurement.
- (2) **We present the first data-driven study analyzing the impact of UI complexity on state-of-the-art LVLMs:** Our baseline analysis uses the accuracy of decisions regarding the next move in the game to identify bottlenecks in performance

and expose the limitations of current models when faced with complex visual applications.

2 Related Work

Games have long served as controlled testbeds for AI, from board and card games [Moravčík et al. 2017; Silver et al. 2018] to sandboxes [Cobbe et al. 2020], due to their fixed rules and discrete actions that facilitate ground-truth labeling. Recent benchmarks extend this to LVLMs by testing spatial reasoning and action planning [Zhang et al. 2024]. However, these primarily assess overall task performance without isolating the impact of the interface itself. This is a critical gap, as recent studies have shown that even state-of-the-art models like GPT-4V require specialized prompting techniques to achieve reliable visual grounding on complex images [Yang et al. 2023], suggesting that inherent UI structure can be a significant challenge.

Drawing from human-computer interaction theories that link visual clutter to cognitive load [Palmer 2003], our work is the first to systematically quantify game UI complexity for LVLM evaluation. While prior work has noted the effect of clutter on human players [Delmas et al. 2022], we propose a new quantifiable score that allows us to directly analyze how UI complexity affects the one-step decision accuracy of LVLMs.

3 Methods

3.1 Dataset Construction

We construct a novel benchmark dataset covering four representative games: *2048*, *Sudoku*, *Governor of Poker*, and *Mahjong Soul*. Unlike existing LVLM benchmarks that mainly focus on natural images or simple visual QA, our benchmark specifically targets interactive game interfaces, which span a wide range of UI complexity. We have ensured that the benchmark provides a controlled but diverse evaluation setting. This enables us to systematically study LVLMs under varying levels of interface complexity.

As a representative game with low visual complexity, we first include *2048*¹, a grid-based puzzle where the player merges tiles by swiping in four possible directions. Ground-truth actions are obtained using an open-source AI solver². Next, *Sudoku*³ is a logic puzzle with fixed solution rules. Instead of solving the entire grid, we focus on predicting the next number in a highlighted cell, ensuring a unique and verifiable ground truth.

To represent higher UI complexity games, we include *Governor of Poker*⁴, which simulates Texas Hold'em with discrete strategic actions such as *Fold*, *Call*, and *All-in*. Ground-truth decisions are supported by a Poker Odds Calculator⁵. Finally, *Mahjong Soul*⁶ features both rich visual elements and decision space. We generate ground-truth strategies using Maka, the official auxiliary tool provided by Mahjong Soul.

For each game, we constructed a dataset of 50 curated screenshots that capture diverse in-game states and decision contexts.

¹<https://play2048.co/>

²<https://lbesson.bitbucket.io/2048-AI/>

³<https://www.websudoku.com/>

⁴<https://www.governorofpoker.com/>

⁵<https://www.pokernews.com/poker-tools/poker-odds-calculator.htm>

⁶<https://mahjong soul.com/>

Table 1: The UI complexity of each game across three key dimensions and the final Score.

	2048	Sudoku	Governor of Poker	Mahjong Soul
Interactive elements (I)	1	1	22-24	8-21
Informational elements (F)	2	3	20-23	18-19
Decision space (D)	4	9	3	2-15
UI Complexity (C)	4.6	6.3	43.4-48.4	27.1-42.8

The number 50 was chosen as a balance between coverage and feasibility: it is large enough to include representative variations of game states, yet sufficiently compact to ensure consistent annotation and manageable evaluation across multiple LVLMs. We created the screenshots using the following strategy:

- Full interfaces were preserved except for the player’s name, in order to reflect realistic playing conditions.
- Screenshots cover different phases of play, e.g., early vs. late boards in *Sudoku*, initial vs. terminal hands in *Mahjong Soul*.
- Screenshots capture various decision points, e.g., *Fold*, *Call*, *Check*, and *All-in* in *Governor of Poker*.

In summary, our dataset ranges from simple grid-based puzzles to visually complex multiplayer games, establishing a pioneering benchmark that systematically aligns LVLM performance with UI complexity. This benchmark forms the foundation for the subsequent experiments and evaluations presented in this paper.

3.2 UI Complexity Definition and Quantification

We define the UI complexity of each game state along three complementary dimensions:

- **Interactive elements (I):** components that the player can directly manipulate, such as buttons, tiles, or cards.
- **Informational elements (F):** regions that display state information, such as scores, timers, hints, or history logs.
- **Decision space (D):** the number of legal actions available in the current state. For instance, *Sudoku* depends on the set of candidate numbers, which are 1-9.

The quantities I and F are counted through manual inspection of interface components, and D is derived from the game rules and current state. The overall complexity score C is defined as:

$$C = \alpha I + \beta F + \gamma \ln(1 + D),$$

where α, β, γ are weights (default $\alpha = \beta = \gamma = 1$). The logarithm term is used to normalize the scale of D , preventing large decision spaces from dominating the complexity measure. Consider the *Sudoku* task where the model is required to predict the next number in a highlighted cell. In this case, the highlighted cell itself is the only interactive element, thus $I = 1$. The board contains three categories of informational cues: originally given digits, blank cells, and numbers already filled by the player, hence $F = 3$. The decision corresponds to choosing a digit between 1 and 9, so $D = 9$. The complexity can be computed as $C = 1 + 3 + \log(10)$, and results in $C \approx 6.3$. Table 1 shows the UI complexity of each game.

The above procedure produces a benchmark dataset which is annotated with complexity scores across three dimensions (I , F ,

and D). Therefore, the dataset enables the systematic evaluation of varying degrees of UI complexity.

3.3 Visual Ability Evaluation

We evaluate LVLMs on the produced dataset using a unified single-step decision task. Given a screenshot and a textual task description, the model is asked to output the optimal action for the current state. The evaluation follows three steps:

- **Prompt design:** Each game is paired with a fixed natural-language instruction. For example, for *2048*, the prompt is “You are playing 2048. Please choose a direction to move.”
- **Model inference:** We query mainstream LVLMs (e.g., GPT-4o [Hurst et al. 2024]) with both the screenshot and the task instruction. External search or reasoning tools are disabled to ensure fair comparison.
- **Answer parsing and scoring:** The model’s response is mapped to a standard action set (e.g., *kong*, or *play 9 bamboo* for mahjong). Accuracy is then computed by comparing predictions with the ground-truth actions.

This setup allows us to present the accuracy separately for each of the four games and analyze how the performance varies with increasing UI complexity.

4 Experiments

4.1 Experimental Setup

We collect a dataset of four games and 200 screenshots, each paired with annotated ground-truth actions, interactive elements, informational elements, and decision space. The screenshots are selected to cover different phases of play and various decision points. Ground-truth labels are obtained using deterministic solvers or well-established domain tools, ensuring that each query has a unique and clear answer.

We benchmark four state-of-the-art LVLMs: GPT-4o [Hurst et al. 2024], Claude-Sonnet-4-0 [Anthropic. 2025], Gemini-2.5-Flash [Team et al. 2023], and Qwen-VL-Max [Bai et al. 2023], accessed via their official APIs. To ensure fairness, all models are prompted with concise, game-specific instructions. For example, *Sudoku* queries are phrased as: “This is a *Sudoku* game. What number should be placed in the cell marked with a red square? Please respond with only the number ‘1-9’.” The prompts are minimal but specific to the task, avoiding unnecessary context. Each screenshot image is queried three times with independent sampling, and the final accuracy is computed as the average across the three runs. A prediction is considered correct if it exactly matches or is semantically equivalent to the ground-truth action. This setup reflects a single-step decision-making setting, where the model outputs one action given a single screenshot, rather than a multi-turn gameplay simulation.

4.2 Experimental Results Analysis

The experimental results are reported in Table 2. Overall, the accuracy across all four games remains modest, reflecting the inherent difficulty of aligning visual UI perception with task-specific reasoning. Among the games, *Mahjong Soul* exhibits the lowest accuracy, from 8.67% to 15.33%, which may be attributed to both its high UI complexity and the requirement to handle hidden information.

Table 2: Average Accuracy (%) of LVLMS' performance with our benchmark.

Model	2048	Sudoku	Governor of Poker	Mahjong Soul
Claude-Sonnet-4.0	27.33%	30.67%	69.33%	10.00%
Gemini-2.5-Flash	38.67%	78.67%	76.67%	15.33%
GPT-4o	29.33%	29.33%	57.33%	12.00%
Qwen-VL-Max	24.00%	42.00%	54.00%	8.67%

Beyond raw accuracy, observations during the experiments reveal meaningful patterns. In *Sudoku* and *2048*, although accuracy is not particularly high (e.g., 27.33% and 30.67% for Claude), the models generally attend to the relevant parts of the interface. Errors in these cases seem to stem primarily from failures in logical computation rather than from misreading the UI itself. In contrast, for *Governor of Poker* and *Mahjong Soul*, we observed frequent outputs that directly copied irrelevant UI elements. For example, in poker the models occasionally returned an interface button instead of a valid action, while in *Mahjong Soul*, some outputs corresponded to options in the left-side menu (e.g., the button of automatic sorting of tiles). These mistakes suggest that, under high UI complexity, the models sometimes struggle to distinguish between visually salient but contextually irrelevant interface components and the actual decision space.

From Table 2, it is worth noting that Gemini consistently achieved higher accuracy than other LVLMS across all games. One possible reason for this is that instruction tuning on heterogeneous multimodal data may give Gemini a greater ability to map task instructions to interface elements, reducing its tendency to be distracted by irrelevant UI components. Additionally, reinforcement learning from human feedback may also have played a role in refining the model's sensitivity to task-relevant signals, enabling it to prioritize decision-space elements more effectively.

Taken together, these insights indicate that the proposed benchmark can capture not only quantitative differences in performance but also qualitatively distinct error behaviors across games. This highlights its potential for systematically evaluating how UI design and task demands jointly influence model decision-making.

5 Dataset Availability and Licensing

We acknowledge the copyrights of the respective game developers. Our use of screenshots for non-commercial, academic critique and analysis is based on the principle of fair use. To foster reproducibility, our complete dataset is publicly released on GitHub under a Creative Commons Attribution-NonCommercial 4.0 (CC BY-NC 4.0) license upon publication.

6 Limitations and Future Works

This study has several limitations. First, the four selected games vary not only in UI complexity but also in their underlying decision spaces. As a result, it is difficult to isolate the effect of UI complexity from that of intrinsic game complexity. Future work could address this limitation by focusing on a single game type with UI designs in multiple levels, which can examine the impact of UI complexity more directly.

Another direction is to complement evaluation with finer-grained behavioral metrics, such as attention to relevant interface regions or

the frequency of outputs derived from irrelevant UI elements. Such analyses could provide a deeper insight into how models process visual UIs and how this processing interacts with task demands.

7 Conclusions

In this work, we introduced the first benchmark specifically designed to evaluate Large Vision-Language Models (LVLMS) in terms of UI complexity. By constructing tasks across four popular games, We demonstrated that different interface designs can present unique challenges to understanding models and making decisions.

Although the absolute accuracy across models varied, the benchmark successfully exposed systematic differences in model behavior, such as misinterpreting irrelevant UI elements or struggling with interfaces of high visual density. These results highlight the necessity of UI-focused evaluation beyond traditional vision-language tasks.

We believe that this benchmark lays the groundwork for future studies of human-computer interaction in the LVLMS period. We hope that it will encourage further research into more controlled game genres, richer evaluation metrics, and UI aware model design.

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